

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

INDEX NUMBER

BIOLOGY

9648/02

Paper 2 Core Paper

15 September 2011

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer **one** question.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	12
2	10
3	12
4	12
5	12
6	12
7	10
Section B	
	20
Total	100
Percentage	
Grade	

This document consists of **18** printed pages.



Section A
Answer **all** questions.

- 1** Fig. 1.1 shows a structure found in an animal cell that is crucial in the cell cycle.

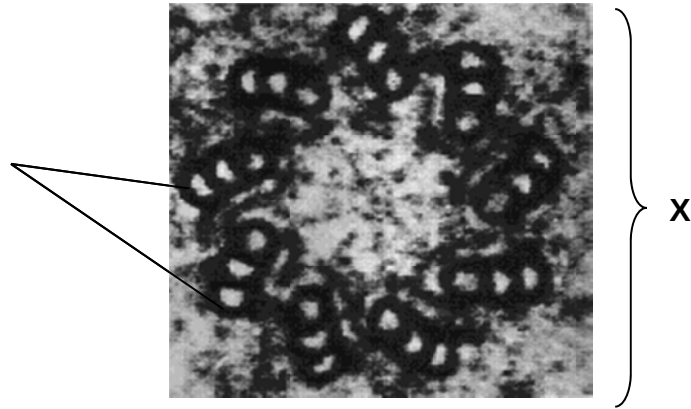


Fig. 1.1

- (a)(i)** Name structure **X**.

..... [1]

- (ii)** Describe arrangement of structure **X**.

.....

 [2]

- (iii)** Explain the role of structure **X** in the M phase of cell cycle.

.....

 [2]

- (b)** Describe the basic unit of structure **Y**.

.....

 [2]

Changes in DNA amount and chromosome numbers were monitored during one cycle of meiosis. Results are shown in Fig. 1.2.

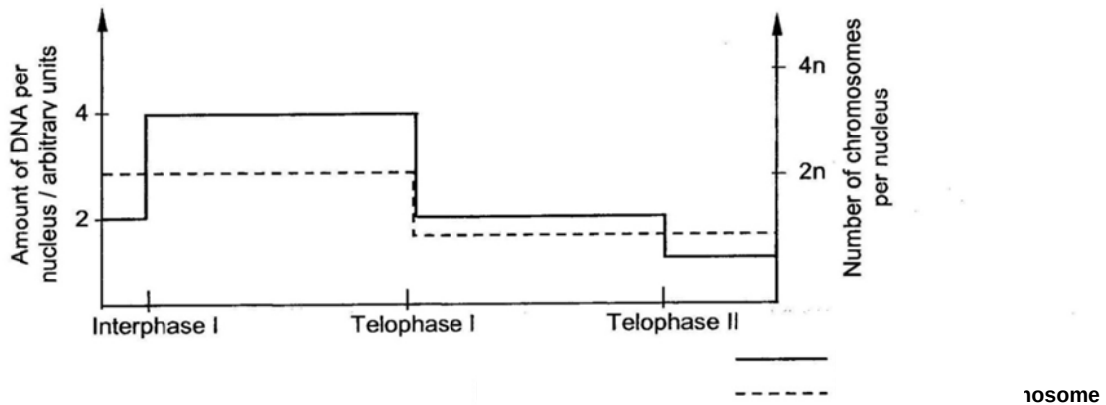


Fig. 1.2

- (c) Explain why DNA amount and chromosome numbers remain constant during the two anaphases in Fig. 1.2.

.....

.....

.....

..... [2]

- (d) Explain the significance of meiosis in genetic variation.

.....

.....

.....

.....

..... [3]

[Total: 12]

- 2 Fig. 2 shows the sequence of bases in a section of the DNA template strand coding for the first seven amino acids of a polypeptide chain.

TACAAGGTCGTCTTTGTCAAA

Fig. 2

- (a)(i) List the corresponding codons in the mRNA transcribed from the template in Fig. 2.

..... [1]

- (ii) Methionine is coded by the initiation codon. On Fig. 2, label, the 3' and 5' end of the DNA template strand. [1]

The polypeptide was hydrolysed to release the first seven amino acids. It contained four different amino acids. The number of each type obtained is shown in the Table 2.

Table 2

Amino acid	Number present
Phe	2
Met	1
Lys	1
Gln	3

- (b) Use the base sequence shown in Fig. 2 to work out the order of amino acids in the polypeptide. [1]

.....

- (c)(i) Explain why single base deletions in the DNA sequence is more likely to produce non-functional proteins than single base substitutions.

.....

.....

.....

..... [2]

- (ii) Describe the effect of a transition mutation at base position 6.

----- [2]

- (d) Explain how the structure of the ribosome is suited to perform its function.

----- [3]

[Total: 10]

- 3 Fig. 3.1 shows a bacterium undergoing the process of binary fission to produce clones.

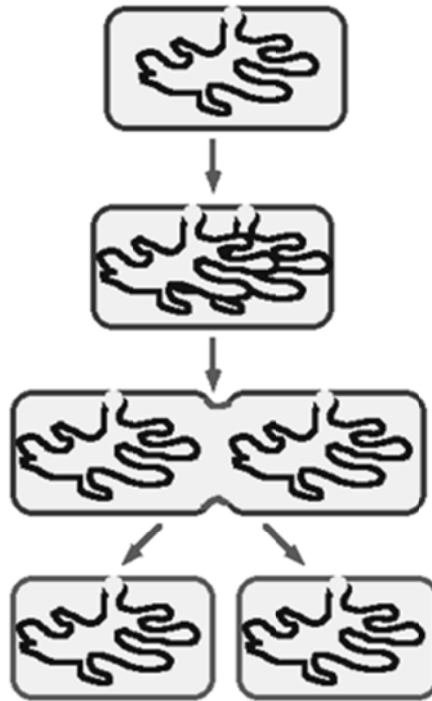


Fig. 3.1

- (a) Explain why these newly synthesized bacteria are described as *clones*.

.....
 [1]

- (b) Describe how binary fission in bacteria brings about genetic stability.

.....

 [2]

While binary fission ensures genetic stability, genetic variation can still occur in bacteria through genetic exchange and recombination. A process that brings about variation is conjugation.

- (c) Describe how genetic variation is brought about through conjugation.

.....

.....

.....

..... [2]

Fig. 3.2 shows a molecular structure of a *lac* repressor that regulates the activity of *lac* operon in a bacterium such as *Escherichia coli*.

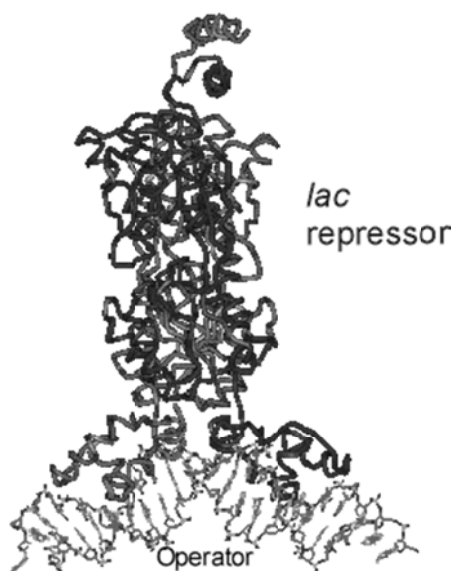


Fig. 3.2

A strain of the bacterium, *Escherichia coli*, has been discovered that has a mutation in the regulator gene. One aspect of the phenotype of this mutant is that it produces large quantities of β -galactosidase at all times.

- (d) Explain why a mutation in the regulator gene leads to the constant production of β -galactosidase.

.....

.....

.....

.....

..... [3]

A certain bacteriophage is known to insert its own genome between the *lacZ* and *lacY* structural genes to give prophages.

(e) Explain the change in number of bacteriophage (if any) in conditions when host bacteria are grown in environment with

(i) abundant supply of glucose and no lactose supply;

.....

.....

.....

..... [2]

(ii) no supply of glucose and abundant supply of lactose.

.....

.....

.....

..... [2]

[Total: 12]

- 4 Cancer cells exhibit a set of unique properties that distinguish them from their normal counterparts. Cancer cells frequently exhibit malfunction in cell cycle checkpoints. One such example is the regulation of passage from G₁ to S phase.

A series of regulatory interactions involving E2F family of transcription factors and Rb proteins are involved in the G₁ to S phase progression. E2F up-regulates transcription of proto-oncogenes whose products are required for passage from G₁ to S phase. The Rb protein represses E2F activity through protein-protein interactions.

The binding of Rb protein to E2F is regulated by Rb's phosphorylation status. Hypophosphorylated Rb binds to E2F while hyperphosphorylated Rb is released from this interaction.

- (a) State **two** properties of cancer cells that distinguish them from their normal counterparts.

[2]

- (b) Explain how a mutation in the Rb gene affecting the phosphorylation status of Rb can result in the development of cancer.

[3]

- (c) Some cancer cells also exhibit mutations in genes coding for histone deacetylase. Suggest how such mutations may lead to cancerous growth.

[2]

- (d) With reference to transcriptional control, describe how the influence of E2F on the expression of proto-oncogenes can be balanced in a eukaryotic cell.

.....

.....

.....

.....

.....

..... [3]

- (e) Explain why cancer usually occurs in older individuals.

.....

.....

.....

..... [2]

[Total: 12]

- 5 (a) Explain what is meant by *homologous pairs of chromosomes*.

.....

.....

.....

..... [2]

An inbred variety of maize, **A**, with finely striped leaves was found to be resistant to the fungus that causes the disease, corn leaf blight.

Plants of variety **A** were crossed with another inbred variety of maize, **B**, which had entirely green leaves and low resistance to the fungus. All the F_1 generation had entirely green leaves and low resistance.

- (b)(i) Using appropriate symbols and indicating what they represent, state the genotype of **A**.

.....

..... [1]

- (ii) The above F_1 generation was back-crossed to variety **A**. Using a genetic diagram, show the expected phenotype of the F_2 generation from this cross.

[3]

The above was carried out (F_1 generation back-crossed to variety **A**) and yielded the following results:

finely striped leaves and resistance	80
finely striped leaves and low resistance	20
entirely green leaves and resistance	22
entirely green leaves and low resistance	78

The χ^2 (chi square) test was done on these data, giving a calculated value of $\chi^2 = 67.36$.

degree of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.31	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Table 5.1

- (c)(i)** Use the calculated value of χ^2 and the table of probabilities provided in Table 5.1 to find the probability of the results of the cross departing significantly by chance from the expected ratio.

Probability: [1]

- (ii)** State what conclusions may be drawn from the probability found in **(c)(i)**.

.....

 [2]

- (iii)** Explain the discrepancy between the actual and expected results.

.....

 [3]

[Total: 12]

- 6 Fig.6.1 shows how a stimulating electrode was used to change the potential difference across an axon membrane. Two other electrodes, **P** and **Q**, were used to record any potential difference produced after stimulation.

The experiment was repeated six times, using a different stimulus potential each time. In experiments **1** to **4**, the stimulating voltage made the inside of the axon less negative. In experiments **5** and **6**, it made the inside of the axon more negative.

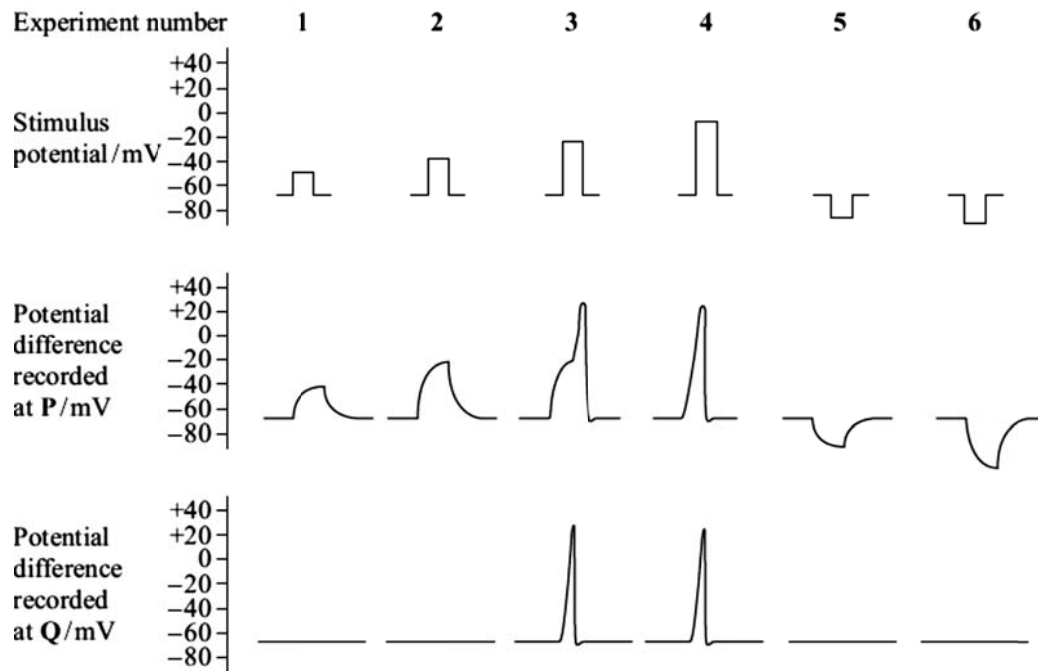


Fig. 6.1

- (a) Explain the results of experiments **2** and **3**.

(i) **2**

.....

.....

..... [2]

(ii) **3**

.....

.....

..... [2]

Myasthenia gravis is a disease which causes muscular weakness. It develops because of an attack by the body's own immune system on neuromuscular junctions. Fig.6.2 below shows a normal neuromuscular junction (nerve to muscle) and one affected by the disease (myasthenic).

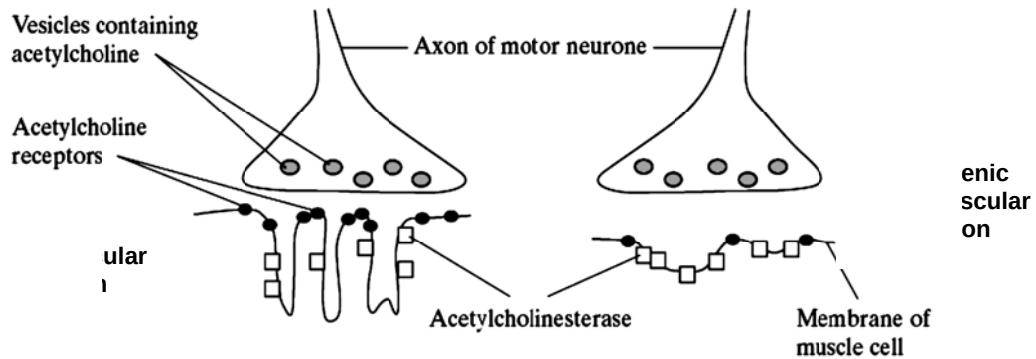


Fig. 6.2

- (b) Describe **two** ways in which a myasthenic neuromuscular junction differs from a normal one and explain how each difference would affect transmissions across the myasthenic neuromuscular junction.

(i) Difference 1:

Effect:

[2]

(ii) Difference 2:

Effect:

[2]

Multiple sclerosis is a disease in which the myelin sheath surrounding neurones are destroyed and so the neurones become de-myelinated. People with multiple sclerosis produce slow responses to stimuli.

- (c) Explain how myelination of neurones helps in speeding up responses.

[2]

The black widow spider venom contains a toxin called latrotoxin that affects the normal functioning of the synapse. The bite of the black widow spider initially produces an increase in neuromuscular activity that leads to painful skeletal muscle spasms and rigidity. This phase of hyperexcitability is rapidly followed by progressive failure of neuromuscular transmission and paralysis. Experiments have found that latrotoxin can form permeable calcium ion pores on the presynaptic membrane.

- (d) Suggest what happens at the synapse of a person bitten by a black widow spider.

.....

.....

.....

..... [2]

[Total: 12]

- 7 Finches inhabiting the Galapagos islands are thought to have evolved from a population of a single species which managed to reach the islands thousands of years ago from South America. A phylogenetic tree was constructed by comparing base sequences in the DNA. Fig. 7.1 shows the evolutionary relationship.

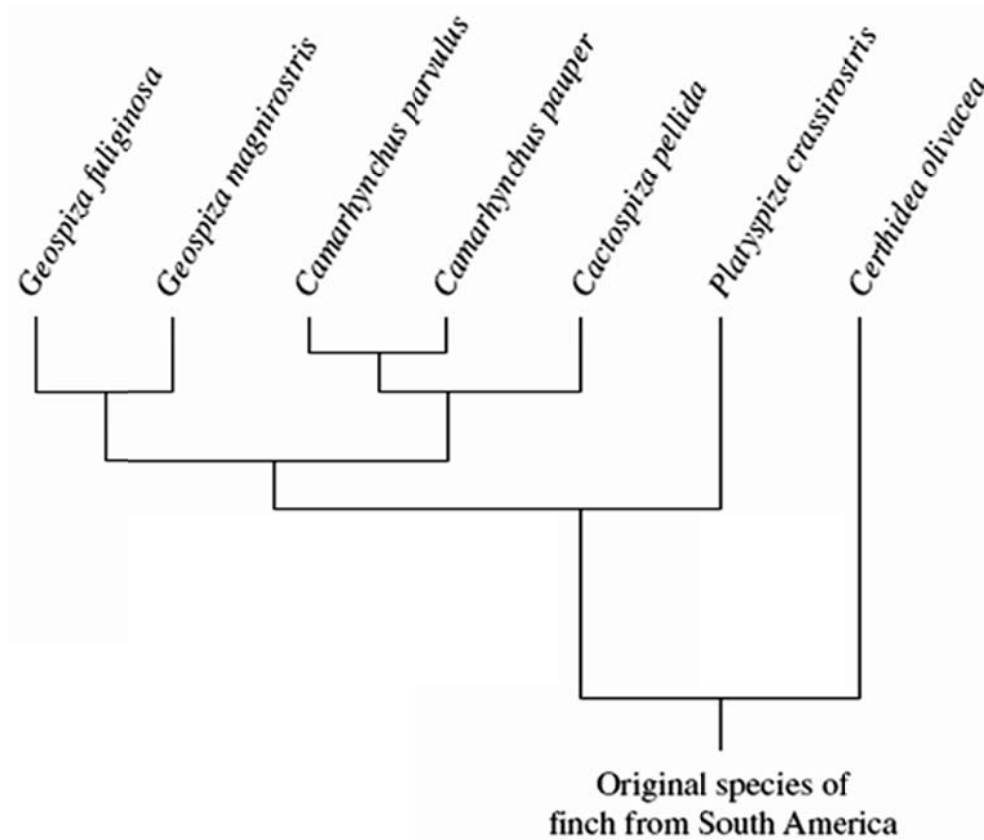


Fig. 7.1

- (a) With reference to Fig. 7.1, state
- (i) which present-day species of finch evolved first on the Galapagos islands;
..... [1]
 - (ii) the number of genera that are present.
..... [1]
 - (iii) the two most closely related genera.
..... [1]
- (b)(i) State **one** other type of evidence that can be used to determine evolutionary relationships.
..... [1]
- (ii) State **one** advantage of using DNA base sequences to determine evolutionary relationships.
..... [1]
- (iii) Explain why other forms of evidence are needed to determine evolutionary relationships despite the advantage of using DNA sequences.
..... [2]

Two species of finches from the *Geospiza* genus living on different islands in the Galapagos was observed to have different beak shapes. Both species are seed eaters. Fig. 7.2 shows the beak morphology of the two species.



Geospiza fuliginosa



Geospiza magnirostris

Fig. 7.2

- (c) With reference to Darwin's theory of natural selection, explain the difference in beak morphology of the two finch species.

----- [3]

[Total: 10]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section (a), (b) etc., as indicated in the question.

- 8 (a) Compare and contrast the ways in which lambda phage and human immunodeficiency virus (HIV) reproduce themselves. [8]
- (b) With reference to a bacteriophage and its reproductive cycle, outline how generalized transduction and transformation could result in a gene transfer between bacteria. [8]
- (c) Some bacteriophages undergo both lysogenic and lytic cycles.

Suggest reasons why it may be advantageous for a bacteriophage to have a lysogenic cycle. [4]

[Total: 20]

- 9** **(a)** Outline the main stages in Krebs cycle. [6]
- (b)** Explain the significance of oxidative phosphorylation in aerobic respiration. [8]
- (c)** With reference to the term 'limiting factor', discuss how changes in light intensity affects the rate of photosynthesis. [6]

[Total: 20]

End of Paper